EXPECTED VALUE AND VARIANCE

Learning Outcome

- calculate expected values, variances, and standard deviations and demonstrate their application to investment problems

The expected value of a random variable is an essential quantitative concept in investments. Investors continually make use of expected values—in estimating the rewards of alternative investments, in forecasting earnings per share (EPS) and other corporate financial variables and ratios, and in assessing any other factor that may affect their financial position. The **expected value of a random variable** is the probability-weighted average of the possible outcomes of the random variable. For a random variable *X*, the expected value of *X* is denoted *E*(*X*).

Expected value (e.g., expected stock return) looks either to the future, as a forecast, or to the "true" value of the mean (the population mean). We should distinguish expected value from the concepts of historical or sample mean. The sample mean also summarizes in a single number a central value. However, the sample mean presents a central value for a particular set of observations as an equally weighted average of those observations. In sum, the contrast is forecast versus historical, or population versus sample.

An equation that summarizes the calculation of the expected value for a discrete random variable X is as follows:

$$E(X) = P(X_1)X_1 + P(X_2)X_2 + \ldots + P(X_n)X_n = \sum_{i=1}^{n} P(X_i)X_i, \qquad 1$$

where X_i is one of *n* possible outcomes of the discrete random variable *X*.

The expected value is our forecast. Because we are discussing random quantities, we cannot count on an individual forecast being realized (although we hope that, on average, forecasts will be accurate). It is important, as a result, to measure the risk we face. Variance and standard deviation measure the dispersion of outcomes around the expected value or forecast.

The **variance of a random variable** is the expected value (the probability-weighted average) of squared deviations from the random variable's expected value:

$$\sigma^2. \qquad 2$$

The two notations for variance are $\sigma^2(X)$ and Var(*X*).

Variance is a number greater than or equal to 0 because it is the sum of squared terms. If variance is 0, there is no dispersion or risk. The outcome is certain, and the quantity *X* is not random at all. Variance greater than 0 indicates dispersion of outcomes. Increasing variance indicates increasing dispersion, all else being equal.

The following equation summarizes the calculation of variance:

$$\sigma^2(X) = P(X_1)[X_1 - E(X)]^2 + P(X_2)[X_2 - E(X)]^2 + \ldots + P(X_n)[X_n - E(X)]^2 = \sum_{i=1}^{n} P(X_i)[X_i - E(X)]^2, \qquad 3$$

where X_i is one of *n* possible outcomes of the discrete random variable *X*.

Variance of *X* is a quantity in the squared units of *X*. For example, if the random variable is return in percent, variance of return is in units of percent squared. Standard deviation is easier to interpret than variance because it is in the same units as the random variable.**Standard deviation** is the square root of variance. If the random variable is return in percent, standard deviation of return is also in units of percent. In the following examples, when the variance of returns is stated as a percent or amount of money, to conserve space, we may suppress showing the unit squared.

The best way to become familiar with these concepts is to work examples.

EXAMPLE 1

BankCorp's Earnings per Share, Part 1

As part of your work as a banking industry analyst, you build models for forecasting earnings per share of the banks you cover. Today you are studying BankCorp. In Exhibit 1, you have recorded a probability distribution for BankCorp's EPS for the current fiscal year.

Exhibit 1: Probability Distribution for BankCorp's EPS

	0.24	2.20
	0.16	2.00
	1.00	

Rate Your Confidence

1. What is the expected value of BankCorp's EPS for the current fiscal year?

Hide Solution

Solution:

Following the definition of expected value, list each outcome, weight it by its probability, and sum the terms.

E(EPS) = 0.15(USD2.60) + 0.45(USD 2.45) + 0.24(USD 2.20) + 0.16(USD 2.00) = USD2.3405

The expected value of EPS is USD2.34.

Category

Probability Trees and Conditional Expectations

2. Using the probability distribution of EPS from Exhibit 1, you want to measure the dispersion around your forecast. What are the variance and standard deviation of BankCorp's EPS for the current fiscal year?

Hide Solution

Solution:

The order of calculation is always expected value, then variance, and then standard deviation. Expected value has already been calculated. Following the previous definition of variance, calculate the deviation of each outcome from the mean or expected value, square each deviation, weight (multiply) each squared deviation by its probability of occurrence, and then sum these terms.

$$\begin{aligned}\sigma^2(\text{EPS}) &= P(2.60)[2.60 - E(\text{EPS})]^2 + P(2.45)[2.45 - E(\text{EPS})]^2 \\ &\quad + P(2.20)[2.20 - E(\text{EPS})]^2 + P(2.00)[2.00 - E(\text{EPS})]^2 \\ &= 0.15(2.60 - 2.34)^2 + 0.45(2.45 - 2.34)^2 \\ &\quad + 0.24(2.20 - 2.34)^2 + 0.16(2.00 - 2.34)^2 \\ &= 0.01014 + 0.005445 + 0.004704 + 0.018496 = 0.038785\end{aligned}$$

Standard deviation is the positive square root of 0.038785:

$\sigma(\text{EPS}) = 0.038785^{1/2} = 0.196939$, or approximately 0.20.

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FA I used financial calc and got 0.230 for SD and mean 2.3125, what is wrong? can someone help plz.

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RR When I calculate it with python it tells me that the std is 0.00

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